

I'll create a comprehensive nmap cheat sheet document for you. Since I can't directly generate PDF files, I'll provide you with well-structured content that you can save as a PDF.

NMAP COMPLETE GUIDE & CHEAT SHEET

From Beginner to Practical Pentester

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1. INTRODUCTION TO NMAP

Nmap (Network Mapper) is the world's most popular network scanning and security auditing tool. It's used to:

- Discover hosts on a network
- Identify open ports
- Detect running services and versions
- Determine operating systems
- Find vulnerabilities using scripts

Basic Syntax:

```
nmap [scan_type] [options] [target]
```

2. UNDERSTANDING NETWORK SCANNING

Legal Considerations:

- **ONLY** scan networks you **OWN**

- **ONLY scan systems with WRITTEN PERMISSION**
- **Our lab rule:** Only scan our own VMs (192.168.56.0/24)

Key Concepts:

IP Addresses: Unique identifier for devices on a network

- Example: 192.168.56.101

Ports: Virtual doors for services (0-65535)

- Port 21 = FTP
- Port 22 = SSH
- Port 80 = HTTP
- Port 443 = HTTPS

Subnet Notation: /24 means 256 addresses (192.168.56.0 to 192.168.56.255)

3. HOST DISCOVERY (PING SWEEPS)

Purpose:

Find which devices are alive on a network before scanning them.

Commands:

```
# Basic ping sweep of entire subnet
nmap -sn 192.168.56.0/24

# Ping sweep with verbose output
nmap -sn -v 192.168.56.0/24

# Scan specific range
nmap -sn 192.168.56.100-150

# Scan list of hosts
nmap -sn 192.168.56.1,101,102
```

What It Does:

- Sends ICMP echo requests
- Sends TCP SYN to port 443
- Sends TCP ACK to port 80
- Sends ICMP timestamp requests
- Identifies which hosts respond

Sample Output:

```
Nmap scan report for 192.168.56.101
Host is up (0.087s latency).
MAC Address: 08:00:27:34:1D:39 (Oracle VirtualBox virtual NIC)
```

4. PORT SCANNING BASICS

Port States:

- **Open ■** - Service is listening and ready

- **Closed** ■ - No service listening
- **Filtered** ■ - Firewall blocking (can't determine)
- **Open|Filtered** - Open or filtered (UDP scan result)

Commands:

```
# Basic scan (top 1000 common ports)
nmap 192.168.56.101

# Scan all 65535 ports
nmap -p- 192.168.56.101

# Scan specific ports
nmap -p 21,22,80 192.168.56.101

# Scan port range
nmap -p 1-1000 192.168.56.101

# Scan top N most common ports
nmap --top-ports 100 192.168.56.101
```

5. SERVICE VERSION DETECTION

Purpose:

Identify exact software and version running on each open port. This reveals specific vulnerabilities.

Command:

```
# Basic version detection
nmap -sV 192.168.56.101

# Aggressive version detection (more accurate)
nmap -sV --version-intensity 5 192.168.56.101

# Light version detection (faster)
nmap -sV --version-intensity 2 192.168.56.101
```

Why It Matters:

```
"FTP running" - Interesting
"vsftpd 2.3.4 running" - EXPLOITABLE (backdoor vulnerability!)
```

Real Example from Metasploitable:

```
21/tcp open ftp vsftpd 2.3.4      → CVE-2011-2523 (backdoor)
22/tcp open ssh OpenSSH 4.7p1    → Old, weak crypto
445/tcp open samba 3.X-4.X       → SambaCry vulnerability
```

6. OS DETECTION

Purpose:

Identify the operating system through TCP/IP stack fingerprinting.

Commands:

```
# Basic OS detection (requires sudo)
sudo nmap -O 192.168.56.101

# OS detection with version scan
sudo nmap -O -sV 192.168.56.101
```

```
# Aggressive OS detection
sudo nmap -O --osscan-guess 192.168.56.101
```

How It Works:

Analyzes tiny differences in how OS responds to packets - like identifying someone by their walking style.

7. SCAN TYPES

7.1 TCP SYN Scan (Stealth)

Flag: -sS (requires sudo)

How it works: Never completes full TCP handshake - sends SYN, gets SYN-ACK, sends RST.

```
sudo nmap -sS 192.168.56.101
```

Advantages: Fast, harder to log, default for privileged users

7.2 TCP Connect Scan

Flag: -sT (no sudo needed)

How it works: Completes full TCP handshake (SYN, SYN-ACK, ACK, then RST).

```
nmap -sT 192.168.56.101
```

Advantages: Doesn't need root, works on all systems

Disadvantages: Slower, easily logged

7.3 UDP Scan

Flag: -sU

Why important: Finds services that TCP scans miss (DNS, DHCP, SNMP, NFS).

```
# Scan top 50 UDP ports
sudo nmap -sU --top-ports 50 192.168.56.101
```

```
# Full UDP scan (very slow)
sudo nmap -sU -p- 192.168.56.101
```

UDP vs TCP:

TCP	UDP
Connection-based	Connectionless
Fast to scan	Slow to scan
Reliable delivery	No guarantee
Examples: HTTP, SSH, FTP	Examples: DNS, DHCP, SNMP

8. NSE SCRIPTS (NMAP SCRIPTING ENGINE)

Script Categories:

Category	Purpose	Example
safe	Info gathering, no harm	http-title

Category	Purpose	Example
default	Basic discovery	ssh-hostkey
vuln	Vulnerability detection	smb-vuln-ms17-010
exploit	Active exploitation	http-shellshock
auth	Authentication checks	ftp-anon
brute	Password cracking	http-brute
dos	Denial of service (dangerous!)	http-slowloris

Commands:

```
# Run default scripts (same as -sC)
nmap -sC 192.168.56.101

# Run all vulnerability scripts
nmap --script=vuln 192.168.56.101

# Run specific script
nmap --script=ftp-anon -p 21 192.168.56.101

# Run script category
nmap --script=smb-* -p 139,445 192.168.56.101

# Run multiple scripts with arguments
nmap --script=http-* --script-args=http.useragent="Mozilla/5.0" -p 80 192.168.56.101
```

Real Example:

```
nmap --script=ftp-anon -p 21 192.168.56.101

# Output:
# ftp-anon: Anonymous FTP login allowed (FTP code 230)
```

9. TIMING TEMPLATES

Flag: -T0 through -T5

Template	Speed	Packet Delay	Use Case
-T0	Paranoid	5 minutes	IDS evasion
-T1	Sneaky	15 seconds	Stealth scanning
-T2	Polite	0.4 seconds	Careful scanning
-T3	Normal	1 second	Default
-T4	Aggressive	10ms	Fast scanning (assumes good network)
-T5	Insane	1ms	Maximum speed (might miss ports)

Example:

```
# Fast scan
sudo nmap -sS -T4 192.168.56.101

# Slow stealth scan
sudo nmap -sS -T2 192.168.56.101
```

10. PORT SELECTION OPTIONS

```
# Single port
nmap -p 21 192.168.56.101

# Multiple ports
nmap -p 21,22,23,80,445 192.168.56.101

# Port range
nmap -p 1-1000 192.168.56.101

# All 65535 ports
nmap -p- 192.168.56.101

# Top N most common
nmap --top-ports 50 192.168.56.101

# Fast scan (top 100 ports)
nmap -F 192.168.56.101
```

11. OUTPUT OPTIONS

Save Results to Files:

Flag	Format	File Extension
-oN	Normal (human-readable)	.nmap
-oX	XML (for tools)	.xml
-oG	Grepable (searchable)	.gnmap
-oA	All formats	Creates all three

Examples:

```
# Save normal output
nmap -sV 192.168.56.101 -oN scan_results.txt

# Save in XML format
nmap -sV 192.168.56.101 -oX scan_results.xml

# Save in all formats
nmap -sV 192.168.56.101 -oA full_scan

# Verbose output
nmap -v 192.168.56.101
nmap -vv 192.168.56.101
```

12. PROFESSIONAL SCAN COMMANDS

The Complete Professional Scan:

```
sudo nmap -sS -sV -sC -O -p- -T4 -v 192.168.56.101 -oA full_recon
```

Breakdown:

- sudo - Run with root privileges
- -sS - SYN stealth scan
- -sV - Service version detection
- -sC - Default scripts

- -O - OS detection
- -p - All 65535 ports
- -T4 - Aggressive timing
- -v - Verbose output
- -oA - Save in all formats

Quick Version (Top 1000 ports):

```
sudo nmap -sS -sV -O -T4 192.168.56.101
```

Fast Version (Quick check):

```
nmap -F 192.168.56.101
```

13. REAL-WORLD METASPLOITABLE LAB RESULTS

Our Target: 192.168.56.101

TCP Open Ports Found:

PORT	STATE	SERVICE	VERSION
21/tcp	open	ftp	vsftpd 2.3.4 (BACKDOOR!)
22/tcp	open	ssh	OpenSSH 4.7p1
23/tcp	open	telnet	Linux telnetd
25/tcp	open	smtp	Postfix smtpd
53/tcp	open	domain	ISC BIND 9.4.2
80/tcp	open	http	Apache httpd 2.2.8
111/tcp	open	rpcbind	2
139/tcp	open	netbios-ssn	Samba 3.X-4.X
445/tcp	open	netbios-ssn	Samba 3.X-4.X (SAMBA CRY!)
512/tcp	open	exec	netkit-rsh rexecd
513/tcp	open	login	
514/tcp	open	shell	Netkit rshd
1099/tcp	open	java-rmi	GNU Classpath grmiregistry
1524/tcp	open	bindshell	ROOT BACKDOOR!
2049/tcp	open	nfs	2-4
2121/tcp	open	ftp	ProFTPD 1.3.1
3306/tcp	open	mysql	MySQL 5.0.51a
5432/tcp	open	postgresql	PostgreSQL 8.3.0-8.3.7
5900/tcp	open	vnc	VNC protocol 3.3
6000/tcp	open	X11	(access denied)
6667/tcp	open	irc	UnrealIRCd (BACKDOOR!)
8009/tcp	open	ajp13	Apache Jserv
8180/tcp	open	http	Apache Tomcat/Coyote

UDP Open Ports Found:

53/udp	open	domain
68/udp	open filtered	dhcpc
69/udp	open filtered	tftp
111/udp	open	rpcbind
137/udp	open	netbios-ns
138/udp	open filtered	netbios-dgm
514/udp	open filtered	syslog
2049/udp	open	nfs

14. QUICK REFERENCE CARDS

Essential Commands Card:

```
# HOST DISCOVERY
nmap -sn 192.168.56.0/24
```

```
# Find live hosts
```

```

nmap -sL 192.168.56.0/24          # List targets without scanning

# BASIC SCANNING
nmap 192.168.56.101              # Default scan (top 1000 ports)
nmap -p 21,22,80 192.168.56.101  # Specific ports
nmap -p- 192.168.56.101          # All ports
nmap --top-ports 100 192.168.56.101 # Top 100 ports

# ADVANCED SCANNING
nmap -sV 192.168.56.101          # Version detection
nmap -O 192.168.56.101           # OS detection (need sudo)
nmap -sS 192.168.56.101          # SYN stealth scan
nmap -sU 192.168.56.101          # UDP scan
nmap -A 192.168.56.101           # Aggressive (everything)

# SCRIPTING
nmap -sC 192.168.56.101          # Default scripts
nmap --script=vuln 192.168.56.101 # Vulnerability scan
nmap --script=ftp-anon -p 21 [target] # Specific script

# OUTPUT
nmap -oN output.txt 192.168.56.101 # Save normal output
nmap -oA full 192.168.56.101        # Save all formats
nmap -v 192.168.56.101              # Verbose

```

TROUBLESHOOTING COMMON ISSUES

Issue	Solution
Permission denied	Use sudo for SYN scans, OS detection
Scan too slow	Use -T4, -F, or --top-ports
Ports missed	Use -p- for all ports, -sU for UDP
Stuck scanning	Press Ctrl+C, check target IP
DNS warnings	Ignore for isolated labs (no DNS server)

CREATING YOUR PDF

To save this as a PDF, follow these steps:

Method 1: From Terminal

```

# Copy everything into a file
cat > nmap_cheatsheet.txt
# Paste content, then Ctrl+D

```

Method 2: Use a Markdown Editor

Save as .md file and convert to PDF using pandoc:

```
pandoc nmap_cheatsheet.md -o nmap_cheatsheet.pdf
```

Method 3: Print to PDF

Open in any text editor, press Ctrl+P, select "Save as PDF".

This document covers everything we've practiced. Keep it bookmarked for reference. When you master these commands, we'll move to Metasploit Framework for actual exploitation! ■